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Advanced Database

Week: 9

1

A data warehouse setup for an e-commerce platform collects data from many sources such as website orders, customer profiles, product listings, and payment systems. All this raw data is first moved into a staging area where it is cleaned, checked, and organized. Errors, duplicates, and missing values are fixed at this stage. After cleaning, the data is loaded into the main data warehouse where it is stored in structured tables. These tables are designed to make queries fast and simple. Businesses can use this data to study customer behavior, track sales performance, and identify trends. ETL tools are used to automate the process of extracting, transforming, and loading data. This setup helps companies make better decisions, improve services, and increase profits by using accurate and reliable information.

2

Star and snowflake schemas are two common ways to design a data warehouse. A star schema has one main fact table connected directly to several dimension tables. This design is simple, easy to understand, and provides fast query performance, which makes it ideal for reporting and analysis. In contrast, a snowflake schema breaks dimension tables into smaller related tables. This reduces data duplication and saves storage space. However, it makes the structure more complex and queries slower because more joins are needed. Star schema is best for speed and simplicity, while snowflake schema is useful when data needs to be more detailed and normalized.

3

An ETL process in Python can be simulated using libraries like pandas. First, data is extracted from a source such as a CSV file or database. Next, the data is transformed by cleaning it, removing duplicates, handling missing values, and changing formats where needed. Finally, the clean data is loaded into a new file or database for storage and analysis. This process improves data quality and ensures consistency across the system. Python makes ETL flexible and easy to automate, especially for large datasets. It is very useful in real-world systems where data needs to be processed regularly. ETL helps turn raw data into meaningful information that businesses can use for decision-making.

4

A data warehouse architecture includes data sources, a staging area, ETL processes, and the main warehouse. Data is collected from different systems such as applications, databases, and external sources. It is first sent to the staging area where it is cleaned and prepared. Then ETL tools move the processed data into the data warehouse. The warehouse stores structured and organized data that is easy to query. Users can access this data through reporting tools and dashboards. The purpose of this architecture is to manage large volumes of data, improve data quality, and support business intelligence. It helps organizations understand patterns, monitor performance, and make informed decisions based on reliable data.